

Sorting Quadrilaterals by Their Attributes

Answer Key

Grade 3 — Geometry

Quick Answers

- | | |
|-------------|---|
| 1. 4 | 7. — |
| 2. 1 | 8. 21 tiles |
| 3. 2 | 9. (a) number of trapezoid tiles = 5, — |
| 4. 1 | 10. (a) how many shapes are in the group = 5, — |
| 5. 30 tiles | |
| 6. 2 | |
-

What is verified

7 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 7, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 3 — Geometry

3.G.A – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

6. *If they got 4: counted how many shapes share the attribute instead of naming the shared number*

The Shape Set, named: A is a square, B a rectangle, C a rhombus, D a parallelogram, E a trapezoid, F a kite. Attribute table (the sorting the worksheet asks the student to produce):

	A	B	C	D	E	F
square corners	4	4	0	0	0	0
pairs of parallel sides	2	2	2	2	1	0
four equal sides	yes	no	yes	no	no	no

1. Shape B: how many square corners?

Solution: Shape B is a rectangle. Test each corner with the corner of a card: all four fit exactly, so all four are square corners.

4

2. Shape E: how many pairs of parallel sides?

Solution: Shape E is a trapezoid. The top and bottom sides stay the same distance apart all the way along — one pair. The two slanted sides lean towards each other, so they are not parallel. That gives exactly one pair. 1

3. How many of the six shapes have four equal sides?

Solution: Measure the four sides of each shape. Shape A (square) has four equal sides, and so does shape C (rhombus) — it is a square pushed over, and pushing a shape over does not change how long its sides are. B, D, E and F all have at least two sides of different lengths. That is two shapes. 2

4. How many more pairs of parallel sides does shape A have than shape E?

Solution: Shape A has two pairs (top with bottom, left with right). Shape E has one pair. “How many more” means subtract: $2 - 1 = 1$. 1

5. Tile shop: 9 squares, 12 rectangles, 4 rhombuses, 5 trapezoids. How many tiles in all?

Solution: All four kinds are quadrilaterals, so they all belong to the one group being counted, and the total is a single sum: $9 + 12 + 4 + 5 = 30$. 30 tiles

6. Most of the six shapes share the same number of pairs of parallel sides. What is that number?

Solution: Count the pairs on each shape in turn: A has 2, B has 2, C has 2, D has 2, E has 1, F has 0. The count that comes up most often is 2. Note what is being asked: not how many shapes share it (that is four), but the number they share. 2

7. (a) Circle every shape with four square corners: a square, a rectangle, a rhombus, a parallelogram. (b) Why does the square belong and the rhombus not?

Solution (a) — judged by eye. The square and the rectangle should be circled, and only those two. Checking each corner against the corner of a card: the square’s four corners fit, the rectangle’s four corners fit, the rhombus’s corners are two sharp and two blunt, and the parallelogram’s are the same. Circling the rhombus is the common error and is not credit.

Solution (b) — instructor-judged. A full-credit answer names the attribute that decides the group. Model answer: “The group is about square corners, not about sides. The square has four square corners, so it is in. The rhombus has four equal sides, but its corners are not square corners, so having equal sides is not enough to get in.” An answer that only says the rhombus is tilted or looks different, without naming the corner attribute, is not full credit.

8. How many tiles sold had four square corners (9 squares, 12 rectangles, 4 rhombuses, 5 trapezoids)?

Solution: First decide which kinds belong to the group: squares and rectangles have four square corners; rhombuses and trapezoids do not. Then add only those two counts: $9 + 12 = 21$. Adding all four numbers gives 30, which answers a different question.

21 tiles

9. (a) How many trapezoid tiles were sold? (b) Why can trapezoid tiles not join the four-square-corners group?

Solution (a): Read the count for trapezoid tiles straight from the sales list.

5

Solution (b) — instructor-judged. A full-credit answer names an attribute a trapezoid fails. Model answer: “A trapezoid has only one pair of parallel sides and its corners are not square corners, so it cannot be in a group that needs four square corners.” Saying only that a trapezoid is a different shape, without naming an attribute, is not full credit. Having four sides is not the issue — every shape on this page has four sides.

10. (a) How many of the six shapes have at least one pair of parallel sides? (b) How can shape A be in more than one group at once?

Solution (a): From the attribute table, the pairs-of-parallel-sides row reads 2, 2, 2, 2, 1, 0. Every shape except F has at least one pair, so five shapes qualify. “At least one” includes the shapes with two.

5

Solution (b) — instructor-judged. A full-credit answer says a shape joins every group whose attribute it has, and names two real groups for shape A. Model answer: “Shape A has four square corners, so it is in that group. It also has four equal sides, so it is in that group too. And it has two pairs of parallel sides. A shape does not get only one label — it belongs to every group whose attribute it has.” Naming one group only, or saying a shape can be in just one group, is not full credit.